

1. GAUSSIAN RV: UNCORRELATEDNESS IMPLIES INDEPENDENCE

Show that if two random variables X and Y are jointly normal and are uncorrelated, then they are independent.

Hint: Two random variables X, Y are jointly normal if

$$\text{pdf}(X, Y) = f_{X,Y}(x, y) = \frac{1}{2\pi |\det(C_Z)|^{1/2}} \exp \left[-\frac{1}{2} (Z - E(Z))^T C_Z^{-1} (Z - E(Z)) \right],$$

in which $Z = [X, Y]^T$.

1. X and Y are uncorrelated, then $E(XY) = E(X)E(Y)$

Noted $X \sim (\bar{X}, \sigma_X^2)$, $Y \sim (\bar{Y}, \sigma_Y^2)$

$$C_Z = E[(Z - E(Z))(Z - E(Z))^T]$$

$$= \begin{bmatrix} E[(X - \bar{X})^2] & E[(X - \bar{X})(Y - \bar{Y})] \\ E[(Y - \bar{Y})(X - \bar{X})] & E[(Y - \bar{Y})^2] \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_X^2 & E(XY) - 2\bar{X}\bar{Y} + \bar{X}\bar{Y} \\ E(XY) - 2\bar{X}\bar{Y} + \bar{X}\bar{Y} & \sigma_Y^2 \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_X^2 & \bar{X}\bar{Y} - 2\bar{X}\bar{Y} + \bar{X}\bar{Y} \\ \bar{X}\bar{Y} - 2\bar{X}\bar{Y} + \bar{X}\bar{Y} & \sigma_Y^2 \end{bmatrix} = \begin{bmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{bmatrix}$$

$$C_Z^{-1} = \begin{bmatrix} \frac{1}{\sigma_X^2} & 0 \\ 0 & \frac{1}{\sigma_Y^2} \end{bmatrix}, |\det(C_Z)|^{1/2} = \sigma_X \sigma_Y$$

$$\text{pdf}(X, Y) = f_{X,Y}(x, y) = \frac{1}{2\pi |\det(C_Z)|^{1/2}} \exp \left[-\frac{1}{2} (Z - E(Z))^T C_Z^{-1} (Z - E(Z)) \right]$$

$$= \frac{1}{2\pi \sigma_X \sigma_Y} \exp \left[-\frac{1}{2} [X - \bar{X}, Y - \bar{Y}] \begin{bmatrix} 1/\sigma_X^2 & 0 \\ 0 & 1/\sigma_Y^2 \end{bmatrix} \begin{bmatrix} X - \bar{X} \\ Y - \bar{Y} \end{bmatrix} \right]$$

$$= \frac{1}{2\pi \sigma_X \sigma_Y} \exp \left[-\frac{1}{2} \left((X - \bar{X})^2 / \sigma_X^2 + (Y - \bar{Y})^2 / \sigma_Y^2 \right) \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_x} \exp\left[-\frac{1}{2}(x-\bar{x})^2/\sigma_x^2\right] \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma_y} \exp\left[-\frac{1}{2}(y-\bar{y})^2/\sigma_y^2\right]$$

$$= f_x(x) \cdot f_y(y)$$

Then we proved that x, y are independent if x, y are jointly normal and are uncorrelated.

2. WIDE-SENSE STATIONARY

Consider the signal plus noise $z(n) = s + v(n)$, where s is a RV with $E[s] = 1$, $E[s^2] = 2$, and for each value of n , $v(n) \sim \mathcal{N}(0, 1)$. It is known that $E[sv(i)] = 1$ for all i , and $v(i)$ is independent of $v(j)$ for all $i \neq j$.

- (1) Compute the autocorrelation function $R_z(i, j)$ of $z(n)$ for all integers i, j .
- (2) Is $z(n)$ WSS (Wide sense stationary)? If so, derive a mathematical expression for $R_z(k)$.

$$2. R_z(i, j) = E[z(i)z(j)]$$

$$E(v^2(n)) = \sigma_{v(n)}^2 + [E(v(n))]^2 = 1$$

$$\textcircled{1} i=j \quad R_z(i, j) = R_z(i, i) = E(z_n^2) = E(s^2 + 2s \cdot v(n) + v^2(n))$$

$$= E(s^2) + 2E(s) \cdot E(v(n)) + E(v^2(n))$$

$$= 2 + 2 \times 1 \times 0 + 1 = 3$$

$$\textcircled{2} i \neq j \quad R_z(i, j) = E[z(i)z(j)] = E[(s+v(i))(s+v(j))]$$

$$= E[s^2 + v(i)s + v(j)s + v(i)v(j)]$$

$$= E(s^2) + E(s)[E(v(i)) + E(v(j))] + E(v(i))E(v(j))$$

$$= 2 + 0 + 0 = 2$$

$$\text{Then } R_z = \begin{bmatrix} 3 & 2 & \dots & 2 \\ 2 & 3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 2 \\ 2 & \dots & 2 & 3 \end{bmatrix}$$

(2) $z(n)$ is WSS.

$$E(z(n)) = E(s+v(n)) = 1+0=1$$

$$R_z(k) = E(z(n)z(n+k)) = \begin{cases} 3, & k=0 \\ 2, & k \neq 0 \end{cases}$$

The upper formula shows that $R_z(k)$ is a function of the time difference $n+k-n=k$, and then $z(n)$ is wide sense stationary.

HOMEWORK 2

1. CASINO EXAMPLE

Assume the transition matrix is

$$A = \begin{bmatrix} & F & L \\ F & 0.95 & 0.05 \\ L & 0.1 & 0.9 \end{bmatrix}$$

and the emission probability matrix is

$$B = \begin{bmatrix} & 1 & 2 & 3 & 4 & 5 & 6 \\ F & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ L & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{2} \end{bmatrix}$$

in which "F" denotes fair die and "L" represents loaded die. Denote Y as the number of dies, and X the status of the die, i.e., $X = 0$ means loaded and $X = 1$ indicates fair. If we have the observation $Y = \{6, 6, 6\}$, use maximum likelihood and maximum *a posteriori* estimate to estimate the status of the die.

1. ① maximum likelihood estimate

$$\begin{aligned} P(Y=(6,6,6) | X=(0,0,0)) &= \left(\frac{1}{2}\right)^3 = \frac{1}{8}, & P(Y=(6,6,6) | X=(0,0,1)) &= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6} = \frac{1}{24} \\ P(Y=(6,6,6) | X=(0,1,0)) &= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{2} = \frac{1}{24}, & P(Y=(6,6,6) | X=(0,1,1)) &= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{72} \\ P(Y=(6,6,6) | X=(1,0,0)) &= \frac{1}{6} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{24}, & P(Y=(6,6,6) | X=(1,0,1)) &= \frac{1}{6} \times \frac{1}{2} \times \frac{1}{6} = \frac{1}{72} \\ P(Y=(6,6,6) | X=(1,1,0)) &= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{2} = \frac{1}{72}, & P(Y=(6,6,6) | X=(1,1,1)) &= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216} \end{aligned}$$

$X=(0,0,0)$ maximizes the likelihood function, $\hat{X}_{ML} = (0,0,0)$

② Maximum a posteriori Estimate

suppose $p(X=0) = \frac{1}{3}$, $p(X=1) = \frac{2}{3}$

$$\begin{aligned} P(X=(0,0)) &= \frac{1}{3} \times 0.9, & P(X=(0,1)) &= \frac{1}{3} \times 0.1, & P(X=(1,0)) &= \frac{2}{3} \times 0.05, & P(X=(1,1)) &= \frac{2}{3} \times 0.95 \\ P(X=(0,0,0)) &= \frac{1}{3} \times 0.9^3 = \frac{27}{100}, & P(X=(0,0,1)) &= \frac{1}{3} \times 0.9 \times 0.1 = \frac{3}{100} \\ P(X=(0,1,0)) &= \frac{1}{3} \times 0.1 \times 0.05 = \frac{1}{600}, & P(X=(0,1,1)) &= \frac{1}{3} \times 0.1 \times 0.95 = \frac{19}{600} \\ P(X=(1,0,0)) &= \frac{2}{3} \times 0.05 \times 0.9 = \frac{3}{100}, & P(X=(1,0,1)) &= \frac{2}{3} \times 0.05 \times 0.1 = \frac{1}{300} \\ P(X=(1,1,0)) &= \frac{2}{3} \times 0.95 \times 0.05 = \frac{19}{600}, & P(X=(1,1,1)) &= \frac{2}{3} \times 0.95 \times 0.95 = \frac{361}{600} \end{aligned}$$

By using Bayes' formula

$$P(X=(0,0,0) | Y=(6,6,6)) = \frac{\frac{1}{3} \times \frac{27}{100}}{P(Y=(6,6,6))}, \quad P(X=(0,0,1) | Y=(6,6,6)) = \frac{\frac{1}{24} \times \frac{3}{100}}{P(Y=(6,6,6))}$$

$$P(X=(0,1,0)|Y=(6,6,6)) = \frac{\frac{1}{24} \times \frac{1}{600}}{P(Y=(6,6,6))}, \quad P(X=(0,1,1)|Y=(6,6,6)) = \frac{\frac{1}{72} \times \frac{19}{600}}{P(Y=(6,6,6))}$$

$$P(X=(1,0,0)|Y=(6,6,6)) = \frac{\frac{1}{24} \times \frac{3}{100}}{P(Y=(6,6,6))}, \quad P(X=(1,0,1)|Y=(6,6,6)) = \frac{\frac{1}{72} \times \frac{1}{300}}{P(Y=(6,6,6))}$$

$$P(X=(1,1,0)|Y=(6,6,6)) = \frac{\frac{1}{72} \times \frac{19}{600}}{P(Y=(6,6,6))}, \quad P(X=(1,1,1)|Y=(6,6,6)) = \frac{\frac{1}{216} \times \frac{361}{600}}{P(Y=(6,6,6))}$$

Then $\hat{X}_{\text{MAP}} = (0,0,0)$

2. DIFFERENT ESTIMATES

- (1) Suppose that $\mathbf{z} = \mathbf{s} + \mathbf{v}$, where $\mathbf{s}$ and $\mathbf{v}$ are independent, jointly distributed RVs with $\mathbf{s} \sim \mathcal{N}(\eta, \sigma^2)$ and $\mathbf{v} \sim \mathcal{N}(0, V^2)$.
 - (a) Derive an expression for $E[\mathbf{s}|\mathbf{z} = \mathbf{z}]$.
 - (b) Derive an expression for $E[\mathbf{s}^2|\mathbf{z} = \mathbf{z}]$.
- (2) Suppose that $\mathbf{z} = \mathbf{s} + \mathbf{v}$, where $\mathbf{s}$ and $\mathbf{v}$ are independent, jointly distributed RVs with $\mathbf{s} \sim \mathcal{N}(\eta_s, \sigma_s^2)$ and $\mathbf{v} \sim \mathcal{N}(0, \sigma_v^2)$. Assume we have measurements $\mathbf{z}(1), \dots, \mathbf{z}(n)$,
 - (a) Derive the maximum likelihood estimate for $\mathbf{s}$;
 - (b) Derive the maximum *a posteriori* estimate for $\mathbf{s}$;
 - (c) Derive the minimum mean square estimate for $\mathbf{s}$;
 - (d) Derive the linear minimum mean square estimate for $\mathbf{s}$;
 - (e) Derive the least squares estimate for $\mathbf{s}$ provided measurements $\mathbf{z}(1), \dots, \mathbf{z}(n)$;
 - (f) Suppose at each time k ($k \in \{1, \dots, n\}$), there is a new measurement $\mathbf{z}(k)$, derive the recursive least squares estimate for $\mathbf{s}$. (Assume $\hat{\mathbf{s}}_0 = E(\mathbf{s})$, the initial error covariance is P_0);
 - (g) Compare all these 6 kinds of estimates.

$$2(1) \quad E(\mathbf{z}) = E(\mathbf{s}) = \eta, \quad \text{Var}[\mathbf{z}] = \text{Var}[\mathbf{s}] + \text{Var}[\mathbf{v}] = \sigma^2 + V^2$$

$$f_s(\mathbf{s}|\mathbf{z}=\mathbf{z}) = \frac{f_s(\mathbf{s}) \cdot f_v(\mathbf{z}-\mathbf{s})}{f_z(\mathbf{z})} = \frac{1}{\sqrt{2\pi} \sqrt{\frac{\sigma^2 V^2}{\sigma^2 + V^2}}} \exp \left[-\frac{(\mathbf{s}-\eta - \frac{\sigma^2}{V^2 + \sigma^2}(\mathbf{z}-\eta))^2}{2 \frac{\sigma^2 V^2}{\sigma^2 + V^2}} \right]$$

$$E[\mathbf{s}|\mathbf{z}=\mathbf{z}] = \eta + \frac{\sigma^2}{V^2 + \sigma^2} (\mathbf{z} - \eta)$$

$$E[\mathbf{s}^2|\mathbf{z}=\mathbf{z}] = \int_{-\infty}^{\infty} \mathbf{s}^2 f_s(\mathbf{s}|\mathbf{z}=\mathbf{z}) d\mathbf{s}$$

$$= E[\mathbf{s}|\mathbf{z}=\mathbf{z}]^2 + D[\mathbf{s}|\mathbf{z}=\mathbf{z}]$$

$$= \left[\eta + \frac{\sigma^2}{V^2 + \sigma^2} (\mathbf{z} - \eta) \right]^2 + \frac{\sigma^2 V^2}{\sigma^2 + V^2}$$

2(2)(a) MLE

$$f(z(1), z(2), \dots, z(n) | s=s) = \prod_{i=1}^n f(z(i) | s=s) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_v} e^{-(z(i)-s)^2/2\sigma_v^2}$$

$$\ln f(z(1), z(2), \dots, z(n) | s=s) = \frac{n}{\sqrt{2\pi}\sigma_v} - \frac{1}{2\sigma_v^2} \sum_{i=1}^n (z(i)-s)^2$$

$$\hat{s}_{ML} = \frac{1}{n} [z(1) + z(2) + \dots + z(n)]$$

(b) MAPE

$$f(s=s | z(1), z(2), \dots, z(n)) = \frac{f(z(1), z(2), \dots, z(n) | s=s) \cdot f_s(s)}{f(z(1), z(2), \dots, z(n))}$$

$$\ln f(s=s | z(1), \dots, z(n)) = g(s) = C - \frac{1}{2\sigma_v^2} \sum_{i=1}^n (z(i)-s)^2 - \frac{1}{2\sigma_s^2} (s-\eta_s)^2$$

$$g'(s) = -\frac{1}{2\sigma_v^2} \cdot \sum_{i=1}^n 2(s-z(i)) - \frac{1}{2\sigma_s^2} \cdot 2(s-\eta_s)$$

$$\hat{s}_{MAP} = \frac{1}{n\sigma_s^2 + \sigma_v^2} \{ \sigma_s^2 [z(1) + z(2) + \dots + z(n)] + \sigma_v^2 \eta_s \}$$

(c) MMSE

First, we demonstrate that $s, z(1), z(2), \dots, z(n)$ are jointly gaussian, because the linear combination of $s, z(1), z(2), \dots, z(n)$, are Gaussian,

$$a_0 s + a_1 z(1) + \dots + a_n z(n) = \left(\sum_{i=0}^n a_i \right) s + \sum_{i=1}^n a_i v(i)$$

is Gaussian with mean $\sum_{i=0}^n a_i \eta_s$ and variance $(\sum_{i=0}^n a_i)^2 \sigma_s^2 + \sum_{i=1}^n a_i^2 \sigma_v^2$

Similarly $z(1), \dots, z(n)$ are also jointly Gaussian.

Assume $\mathbf{z} = [z(1), \dots, z(n)]^T$, and as s and $\mathbf{z}$ are jointly Gaussian

$$(s, \mathbf{z}) \sim N \left(\begin{bmatrix} \eta_s \\ \eta_z \end{bmatrix}, \begin{bmatrix} \Sigma_{ss} & \Sigma_{sz} \\ \Sigma_{zs} & \Sigma_{zz} \end{bmatrix} \right) = N(\eta, \Sigma)$$

and then

$$\Sigma = \begin{bmatrix} \Sigma_{ss} & \Sigma_{sz} \\ \Sigma_{zs} & \Sigma_{zz} \end{bmatrix} = \begin{bmatrix} I & \Sigma_{sz} \Sigma_{zz}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma_{ss} - \Sigma_{sz} \Sigma_{zz}^{-1} \Sigma_{zs} & 0 \\ 0 & \Sigma_{zz} \end{bmatrix} \begin{bmatrix} I & 0 \\ \Sigma_{zz}^{-1} \Sigma_{zs} & I \end{bmatrix}$$

$$\Sigma^{-1} = \begin{bmatrix} I & 0 \\ -\Sigma_{zz}^{-1} \Sigma_{zs} & I \end{bmatrix} \begin{bmatrix} (\Sigma_{ss} - \Sigma_{sz} \Sigma_{zz}^{-1} \Sigma_{zs})^{-1} & 0 \\ 0 & \Sigma_{zz}^{-1} \end{bmatrix} \begin{bmatrix} I & -\Sigma_{sz} \Sigma_{zz}^{-1} \\ 0 & I \end{bmatrix}$$

the joint distribution $p(s, z)$

$$p(s, z) = \frac{1}{\sqrt{(2\pi)^{n+1} \det \Sigma}} \exp \left(-\frac{1}{2} (x - \eta)^T \Sigma^{-1} (x - \eta) \right)$$

where $x = [s, z^T]^T$, $\det \Sigma = \det(\Sigma_{zz}) \cdot \det(\Sigma_{ss} - \Sigma_{sz} \Sigma_{zz}^{-1} \Sigma_{zs})$

and the quadratic part is

$$(x - \eta)^T \Sigma^{-1} (x - \eta) = [(s - \eta_s)^T, (z - \eta_z)^T] \Sigma^{-1} \begin{bmatrix} s - \eta_s \\ z - \eta_z \end{bmatrix}$$

$$p(s|z) = \frac{p(s, z)}{p(z)} \sim N(\eta_s + \Sigma_{sz} \Sigma_{zz}^{-1} (z - \eta_z), \Sigma_{ss} - \Sigma_{sz} \Sigma_{zz}^{-1} \Sigma_{zs})$$

the MMSE estimate is

$$\hat{s}_{\text{MMSE}} = E(s|z) = \eta_s + \Sigma_{sz} \Sigma_{zz}^{-1} (z - \eta_z)$$

(d) LMMSE

$$\hat{s}_{\text{LMMSE}} = E(s z^T) [E(z z^T)]^{-1} z$$

where $E(s z^T) = E(s z(1), \dots, s z(n)) = [\eta_s^2 + \sigma_s^2, \dots, \eta_s^2 + \sigma_s^2]$

for example, $E(s z(i)) = E[s(s + v(i))] = E(s^2) + E(s) E(v(i)) = \eta_s^2 + \sigma_s^2$

$$E(z z^T) = \begin{bmatrix} E[z(1)^2] & \dots & E[z(1)z(n)] \\ \vdots & & \vdots \\ E[z(n)z(1)] & \dots & E[z(n)^2] \end{bmatrix}$$

$$\textcircled{1} \text{ if } i \neq j, E[z(i)z(j)] = E[(S+V(i))(S+V(j))] = E(S^2) + 0 = \eta_s^2 + \sigma_s^2$$

$$\textcircled{2} \text{ if } i = j, E[z(i)z(j)] = E[S^2] + E[V(i)^2] = \eta_s^2 + \sigma_s^2 + \sigma_v^2$$

$$E(zz^T) = \begin{bmatrix} \eta_s^2 + \sigma_s^2 + \sigma_v^2 & \eta_s^2 + \sigma_s^2 & \cdots & \eta_s^2 + \sigma_s^2 \\ \eta_s^2 + \sigma_s^2 & \eta_s^2 + \sigma_s^2 + \sigma_v^2 & & \vdots \\ \vdots & & \ddots & \vdots \\ \eta_s^2 + \sigma_s^2 & \cdots & & \eta_s^2 + \sigma_s^2 + \sigma_v^2 \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_v^2 & 0 \\ & \ddots \\ 0 & & \sigma_v^2 \end{bmatrix} + \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} (\eta_s^2 + \sigma_s^2) [1, \cdots, 1]$$

with Sherman-Morrison-Woodbury equations,

$$(A^{-1} + BD^{-1}C)^{-1} \equiv A - AB(D + CAB)^{-1}CA$$

and then

$$[E(zz^T)]^{-1} = \frac{1}{\sigma_v^2 [\sigma_v^2 + n(\sigma_s^2 + \eta_s^2)]} \begin{bmatrix} \sigma_v^2 + (n-1)(\sigma_s^2 + \eta_s^2) & \cdots & -(\sigma_s^2 + \eta_s^2) \\ \vdots & \ddots & \vdots \\ -(\sigma_s^2 + \eta_s^2) & \cdots & \sigma_v^2 + (n-1)(\sigma_s^2 + \eta_s^2) \end{bmatrix}$$

Ultimately, the LMMSE estimate is

$$\hat{S}_{\text{LMMSE}} = \frac{\eta_s^2 + \sigma_s^2}{\sigma_v^2 + n(\sigma_s^2 + \eta_s^2)} \sum_{i=1}^n z(i)$$

(e) LS

$$H = [1, 1, 1, \cdots, 1] \in \mathbb{R}^{n \times 1}$$

$$\hat{S}_{\text{LS}} = (H^T H)^{-1} H^T z = \frac{1}{n} \sum_{i=1}^n z(i)$$

(f) RLS

$$R_k = E(V_k^2) = R = \sigma_v^2, H_k = 1$$

Initial estimate:

$$\hat{S}_0 = E(x)$$

the first measurement ($k=1$):

$$K_1 = P_{k-1} H_k^T (H_k P_{k-1} H_k^T + R_k)^{-1} = P_0 (P_0 + R)^{-1}$$

$$\hat{S}_1 = \hat{S}_{k-1} + K_k (z_k - H_k \hat{S}_{k-1}) = \hat{S}_0 + \frac{P_0}{P_0 + R} (z(1) - \hat{S}_0)$$

$$P_1 = (I - K_k H_k) P_{k-1} (I - K_k H_k)^T + K_k R_k K_k^T = \frac{P_0 R}{P_0 + R}$$

measurement ($k=2$):

$$k_2 = \frac{P_1}{P_1 + R} = \frac{P_0}{2P_0 + R}$$

$$P_2 = \frac{P_1 R}{P_1 + R} = \frac{P_0 R}{2P_0 + R}$$

$$\hat{S}_2 = \hat{S}_1 + \frac{P_1}{P_1 + R} (z(2) - \hat{S}_1) = \frac{P_0 + R}{2P_0 + R} \hat{S}_1 + \frac{P_0}{2P_0 + R} z(2)$$

measurement (k):

$$P_{k-1} = \frac{P_0 R}{(k-1)P_0 + R}$$

$$k_k = \frac{P_0}{kP_0 + R}$$

$$\hat{S}_k = \hat{S}_{k-1} + k_k (z(k) - \hat{S}_{k-1}) = (1 - k_k) \hat{S}_{k-1} + k_k z(k)$$

$$= \frac{(k-1)P_0 + R}{kP_0 + R} \hat{S}_{k-1} + \frac{P_0}{kP_0 + R} z(k)$$

(g) Comparison

$$\hat{S}_{ML} = \frac{1}{n} [z(1) + z(2) + \dots + z(n)]$$

$$\hat{S}_{MAP} = \frac{1}{n\sigma_s^2 + \sigma_v^2} \{ \sigma_s^2 [z(1) + z(2) + \dots + z(n)] + \sigma_v^2 \eta_s \}$$

$$\hat{S}_{MMSE} = E(S|Z) = \eta_s + \Sigma_{S2} \Sigma_{22}^{-1} (Z - \eta_2)$$

$$\hat{S}_{LMMSE} = \frac{\eta_s^2 + \sigma_s^2}{\sigma_v^2 + n(\sigma_s^2 + \eta_s^2)} \sum_{i=1}^n z(i)$$

$$\hat{S}_{LS} = (H^T H)^{-1} H^T Z = \frac{1}{n} \sum_{i=1}^n z(i)$$

① $\hat{S}_{LS} = \hat{S}_{ML}$, and if $\sigma_v^2 \rightarrow 0$, $\hat{S}_{MAP} \rightarrow \hat{S}_{ML}$, $\hat{S}_{LMMSE} \rightarrow \hat{S}_{ML}$

② if s and z are zero-mean, jointly Gaussian, the LMMSE estimate is also the optimal MMSE estimate.

③ Compared to LS, RLS is more computational especially for a scale parameter s and a perfect measurement of it. i.e. $H_i=1, R_i=0$.

(1)

I. Extended Kalman filter

① Initialize $\hat{x}_0 = E(x_0) = [0, 5]^T$ $\hat{p}_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 1 \end{bmatrix}$

② for $k=1$

$$F_0 = \left. \frac{\partial f_0}{\partial x} \right|_{(\hat{x}_0, 0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \bigg|_{(\hat{x}_0, 0)} = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

$$L_0 = \left. \frac{\partial f_0}{\partial w} \right|_{(\hat{x}_0, 0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\check{p}_1 = F_0 \hat{p}_0 F_0^T + L_0 Q_0 L_0^T = \begin{bmatrix} 0.36 & 0.5 \\ 0.5 & 1.1 \end{bmatrix}$$

$$\check{x}_1 = f_0(\hat{x}_0, u_0, 0) = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \cdot (-2) = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix}$$

$$H_1 = \left. \frac{\partial h_1}{\partial x} \right|_{(\check{x}_1, 0)} = [5.4 \times 10^{-3}, 0] \quad M_1 = \left. \frac{\partial h_1}{\partial v} \right|_{(\check{x}_1, 0)} = 1$$

Perform the measurement update:

$$K_1 = \check{p}_1 H_1^T (H_1 \check{p}_1 H_1^T + M_1 R_1 M_1^T)^{-1} = \begin{bmatrix} 0.1944 \\ 0.17 \end{bmatrix}$$

$$\hat{x}_1 = \check{x}_1 + K_1 [y_1 - h_1(\check{x}_1, 0)] = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix} + \begin{bmatrix} 0.1944 \\ 0.17 \end{bmatrix} \times \frac{20 - 19.2}{180} \cdot \pi = \begin{bmatrix} 2.5027 \\ 4.00378 \end{bmatrix}$$

$$\hat{p}_1 = (I - K_1 H_1) \check{p}_1 = \begin{bmatrix} 0.3596 & 0.4995 \\ 0.4995 & 1.0993 \end{bmatrix}$$

II unscented Kalman filter

① Form the matrix of sigma points:

$$\hat{x}_0 = E(x_0) = [0, 5]^T, \hat{p}_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \sqrt{\hat{p}_0} = \begin{bmatrix} 0.1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ choose } \lambda = 1$$

$$\hat{X}_0 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 5 & 5 & 5 & 5 & 5 \end{bmatrix} + \sqrt{\lambda + \lambda} \begin{bmatrix} 0 & 0.1 & 0 & -0.1 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}$$

② propagate the sigma point through the dynamic model.

$$\hat{X}_1 = f(\hat{X}_0, u_0, 0) = \begin{bmatrix} 2.5 & 2.67 & 3.37 & 2.33 & 1.63 \\ 4 & 4 & 5.73 & 4 & 2.26 \end{bmatrix}$$

③ Compute the predicted mean and covariance.

$$\text{Set } W_0^{(m)} = W_0^{(c)} = \frac{1}{5}, W_i^{(m)} = W_i^{(c)} = \frac{1}{6}, i=1, 2, 3, 4, \alpha = \beta = 0$$

$$\check{x}_1 = \sum_{i=1}^4 W_i^{(m)} \hat{X}_{1,i} = [2.5, 4]^T$$

$$\check{p}_1 = \sum_{i=1}^4 W_i^{(c)} (\hat{X}_{1,i} - \check{x}_1)(\hat{X}_{1,i} - \check{x}_1)^T + Q_0 = \begin{bmatrix} 0.36 & 0.5 \\ 0.5 & 1.1 \end{bmatrix}$$

④ update step

$$\check{X}_1 = \begin{bmatrix} 2.5 & 2.5 & 2.5 & 2.5 & 2.5 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix} + \sqrt{5} \begin{bmatrix} 0 & 0.6 & 0.71 & -0.6 & -0.71 \\ 0 & 0.71 & 1.05 & -0.71 & -1.05 \end{bmatrix}$$

$$= \begin{bmatrix} 2.4 & 3.54 & 3.72 & 1.46 & 1.28 \\ 4 & 5.23 & 5.82 & 2.78 & 2.18 \end{bmatrix}$$

⑤ propagate the sigma point through the measurement model:

$$\hat{Y}_{1,i} = h(\check{X}_{1,i}, 0) = [0.33475, 0.3405, 0.3414, 0.3292, 0.3283]$$

compute $\mu_1 = \sum_{i=0}^4 w_i^{(m)} \hat{Y}_{1,i} = 0.33475$

$$S_1 = \sum_i w_i^{(c)} (\hat{Y}_{1,i} - \mu_1) (\hat{Y}_{1,i} - \mu_1)^T + R_1 = 2.5 \times 10^{-5} + 0.05 = 0.050025$$

$$C_1 = \sum_{i=1}^4 w_i^{(c)} (\check{X}_{1,i} - \check{X}_1) (\hat{Y}_{1,i} - \mu_1) = \begin{bmatrix} 4.54 \times 10^{-3} \\ 6.28 \times 10^{-3} \end{bmatrix}$$

$$K_1 = C_1 S_1^{-1} = \begin{bmatrix} 0.09075 \\ 0.11256 \end{bmatrix}$$

$$\hat{X}_1 = \check{X}_1 + K_1 (y_1 - \mu_1) = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix} + \begin{bmatrix} 0.09075 \\ 0.11256 \end{bmatrix} \left(\frac{20}{180} \pi - 0.33475 \right) = \begin{bmatrix} 2.5013 \\ 4.0018 \end{bmatrix}$$

$$\hat{P}_1 = P_1 - K_1 S_1 K_1^T = \begin{bmatrix} 0.36 & 0.5 \\ 0.5 & 1.1 \end{bmatrix} - \begin{bmatrix} 0.09075 \\ 0.11256 \end{bmatrix} 0.050025 \begin{bmatrix} 0.09075 & 0.11256 \end{bmatrix}$$

$$= \begin{bmatrix} 0.3596 & 0.4994 \\ 0.4994 & 1.0992 \end{bmatrix}$$

Comprison: UKT needs more computations than EkF, but the result generated by the two method are similar.

$$(2) \quad F = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \quad H = [1 \ 0] \quad R = 0.05, \quad Q = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}$$

And the steady-state kalman filter exists iff the following conditions holds

① (F, H) is detectable

② (F, J) is stabilizable

① $\text{rank} \begin{bmatrix} F \\ H \end{bmatrix} = \text{rank} \begin{bmatrix} 1 & 0 \\ 1 & 0.5 \end{bmatrix} = 2 \Rightarrow (F, H)$ is detectable.

② $\det(\lambda I - F) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 1$

So the uncontrollable state is unstable, then it is not stabilizable

Therefore, there isn't steady-state kalman filter.